MND CODE

PROJECT SPECIFICATION & ARCHITECTURE

1. Platform Vision & Role

This document serves as the master architectural and design blueprint for building a complete, production-oriented educational website for **MND CODE**, specializing in networking, cloud computing, cybersecurity, and ethical hacking. The objective is to design a platform that surpasses typical training institute websites through a distinctive visual identity, outstanding typography, cinematic transitions, and interactive learning components.

> **Brand Positioning:** A modern technical education platform helping students and aspiring IT professionals build practical skills in computer networking, cloud infrastructure, cybersecurity, and authorized penetration testing.
> **Tagline:** "Connect. Compute. Secure."
> **Supporting Headline:** "Build Your Future in Networking, Cloud & Cybersecurity."

2. Visual Design System

The visual language establishes a futuristic, premium technology aesthetic. It combines modern editorial layouts, subtle glassmorphism, spatial depth, and restrained cyber-inspired visual effects. The interface prioritizes readability through clear type scales and excellent contrast, utilizing **Space Grotesk** for geometric, technical headings and **Inter** for highly legible body copy.

Interactive elements will employ animated gradient lighting, subtle grid backgrounds, and smooth page transitions while respecting reduced-motion preferences.

#070A12 Background	#101625 Surface	#3B82F6 Primary Blue	#22D3EE Electric Cyan	#8B5CF6 Purple Accent	#22C55E Success

3. Technology Stack Architecture

The platform is architected for performance, scalability, and security, paving a clear path toward backend integration while remaining functional as a robust prototype.

- **Frontend:** React with TypeScript, Vite (or Next.js for SSR advantages), Tailwind CSS for utility styling, Framer Motion for cinematic animations, and Lucide icons.
- **Backend-Ready Infrastructure:** Node.js with NestJS or Express, providing a REST API with strict validation and error handling.
- **Data & Storage:** PostgreSQL database for relational data integrity, secure authentication flows, and Cloud object storage for heavy course assets.

4. Complete Website Structure

The platform is divided into three distinct routing domains, sharing a consistent navigation system and strict server-side route protection.

Public Pages

- Home (/)
- About Us (/about)
- All Courses (/courses)
- Networking (/courses/networking)
- Cloud (/courses/cloud-computing)
- Cybersecurity (/courses/ethical-hacking)
- Learning Paths (/learning-paths)
- Practical Labs (/labs)
- Projects & Resources (/projects)
- Blog & FAQ (/blog, /faq)
- Legal & Contact (/privacy, /contact)

Student & Admin Routes

- **Student Login/Register** (/login)
- **Student Dashboard** (/dashboard)
- **My Courses** (/dashboard/courses)
- **Lesson Player** (/learn/:id/:lesson)
- **Progress & Settings**
- **Admin Dashboard** (/admin)
- **Course Management**
- **Student Management**

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5. Homepage Experience

The homepage is designed as a high-impact narrative, guiding users from initial discovery to practical engagement.

Hero Section: Animated Network Nodes & Cloud Architecture	Responsive, lightweight vector animation demonstrating data flow.

- **A. Announcement Bar:** Configurable alerts for new resources.
- **B. Premium Navigation:** Transparent glass header with active indicators.
- **C. Hero Section:** "Master Networks. Build in the Cloud. Secure the Future." Includes dual CTAs to explore courses or practical labs.
- **D. Trust Highlights:** Cards emphasizing structured paths, hands-on exercises, and responsible security practices (avoiding fabricated metrics).
- **E. Featured Courses:** Three premium course cards displaying level, duration, and outcomes.
- **F. Learning Roadmap:** Visual progression from IT Fundamentals → Networking → Cloud → Security.
- **G. Interactive Labs Preview:** Browser-based demonstrations of subnetting challenges or packet-flow diagrams.
- **H. Projects Showcase:** Examples like "Small office network design" and "Authorized web security assessment."
- **I. Why MND CODE:** Bento-grid layout explaining the practical, responsible learning approach.
- **J. Learning Process:** Discover → Learn → Practice → Build → Review.
- **K. FAQ & Final CTA:** Common prerequisites questions ending with "Your Next Chapter Starts Here."

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COURSE 1: Network Setup & Administration

Route: /courses/networking
Audience: Beginners, IT support aspirants, aspiring network administrators.
Goal: Master how devices communicate, network design, configuration, and troubleshooting.

Module 1–3: Fundamentals, OSI & Subnetting

Introduction to LAN/WAN topologies, throughput, and latency. Deep dive into the seven OSI layers, TCP/IP architecture, and packet travel. Extensive practice with IPv4/IPv6, CIDR notation, subnet masks, and binary-to-decimal calculations.

Module 4–6: Routing, Switching & Services

Layer 2/3 concepts, VLANs, trunking, and static/dynamic routing protocols. Network segmentation basics. Understanding DNS resolution, DHCP scopes, NAT/PAT, and Wi-Fi standards (WPA2/WPA3, SSIDs, channel planning).

Module 7–8: Security & Troubleshooting

Firewall fundamentals, ACLs, VPN concepts, and secure SSH administration. Practical troubleshooting using ping, traceroute, ss, and dig. Basic packet capture analysis with Wireshark and incident documentation.

Module 9–10: Design Projects & Assessment

Design a segmented small-office network with a custom IP addressing plan. Final practical configuration tasks, troubleshooting scenarios, and a comprehensive network design review.

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COURSE 2: Cloud Computing & Networking

Route: /courses/cloud-computing
Audience: Networking students, developers, aspiring cloud engineers.
Goal: Understand virtual infrastructure, secure cloud connectivity, and deployment.

Module 1–3: Cloud Basics & Virtual Networks

IaaS, PaaS, SaaS models and the shared responsibility framework. Introduction to core compute and storage services across AWS, Azure, and GCP. Designing Virtual Private Clouds (VPCs), public/private subnets, and internet gateways.

Module 4–6: Connectivity, Security & Compute

Site-to-site VPNs, transit networking, and hybrid cloud connectivity. Cloud network security including security groups, IAM, least privilege, and bastion hosts. Deploying virtual machines, introductory Kubernetes networking, and autoscaling.

Module 7–8: Monitoring & Infrastructure as Code

Analyzing network flow logs, metrics, and routing failures. Introduction to Terraform for reusable, version-controlled cloud infrastructure deployment and safe configuration validation.

Module 9–10: Optimization & Cloud Projects

Cost-aware architecture, resource tagging, and disaster recovery planning. Final project: Deploying a segmented virtual network, configuring a hybrid connection simulation, and producing a formal architecture review.

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COURSE 3: Ethical Hacking & Pen Testing

Route: /courses/ethical-hacking
Audience: Aspiring security professionals and system administrators.
Goal: Learn vulnerability analysis, web security, and responsible assessment reporting.

REQUIRED PRINCIPLE

All hands-on security exercises MUST be restricted to explicitly authorized, intentionally vulnerable lab environments owned by the learner.

Module 1–4: Security Fundamentals & Lab Setup

CIA triad, threat modeling, and ethics/rules of engagement. Linux command-line mastery (permissions, shell fundamentals). Setting up isolated virtual machines and snapshot-recoverable vulnerable targets.

Module 5–7: Reconnaissance, Assessment & Web Sec

Defining scope, passive information gathering, and asset inventories. Using vulnerability scanners in controlled labs (CVSS fundamentals). Analyzing HTTP mechanisms, injection, XSS, and OWASP Top 10 concepts.

Module 8–10: Hardening, Methodology & Reporting

Secure configuration baselines, service minimization, and defensive monitoring. Full penetration testing lifecycle: Pre-engagement authorization, controlled validation, risk classification, and writing executive remediation reports.

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6. Core Platform Features

Beyond static content, the MND CODE platform provides interactive tools that bridge theoretical knowledge with practical execution.

Interactive Learning Labs

Browser-based educational tools designed for safe, instant feedback. These simulations never execute live external scans.

Subnetting Calculator
Dynamic calculation of CIDR, network address, and broadcast ranges.

Packet Journey Visualizer
Animated paths showing encapsulation across OSI layers.

Linux Terminal Simulator
Safe, simulated command-line quizzes for basic administration.

Architecture Builder
Drag-and-drop cloud networking components to evaluate valid connections.

Student Dashboard Experience

A personalized, authenticated hub where learners track their progress. The dashboard features a clean, widget-based layout tracking overall module completion, quiz results, learning streaks, and quick links to resume their most recent video lesson or lab environment.

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